



PASTLE: Pivot-aided space transformation for local explanations

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ABSTRACT

During the last decade, more and more Artificial Intelligence systems have been designed using complex and sophisticated architectures to reach unprecedented predictive performance. The side effect is an increase in opacity of their inner workings which is inadmissible when such systems are applied in critical domains (healthcare, finance and so on). The *eXplainable AI* (XAI) research field aims to overcome this limitation thus helping humans to understand black-box decisions. In this paper we propose a novel model-agnostic XAI technique, named *Pivot-Aided Space Transformation for Local Explanations* (PASTLE), which exploits an instance-space transformation to explain any model's predictions, aiming to enhance human trust towards the AI decisions. We experimentally evaluate the effects of the introduced space transformation on various real-world data sets and our user study reveals promising results in terms of effective explainability.

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1. Introduction

Nowadays, sophisticated machine learning models are assisting humans in the most disparate domains, but they usually are so complex and intricate that no one – neither their designers – is able to understand the “cognitive” processes leading to decisions.

The main reason for the current focus on the *eXplainable AI* (XAI) topic is the increasing number of AI applications, especially in critical domains (e.g. medicine, military, finance) where the interpretability of decisions is becoming a mandatory requirement. According to DARPA [1], while producing more explainable models, it is necessary to maintain a high level of learning performance (e.g. prediction accuracy) and to enable human users to understand, appropriately trust, and effectively manage the emerging generation of artificially intelligent partners. In the quest of reaching these goals, XAI research is divided into two branches: (1) *model-specific* techniques, which deal with the inner workings of specific models, and (2) *model-agnostic* techniques, which aim to design new systems capable of explaining predictions of any existing models. In addition, XAI methodologies differ on the scope of interpretability, which can be *local*, if the explanation refers to an individual instance, or *global*, if it refers to the overall behavior of the model [2]. Most local approaches (e.g. LIME [3], SHAP [4]) capture the importance of features over the explained prediction, presenting

them to the human user in a easy-to-understand manner. Although these explanations are locally accurate, it is not clear how they would apply to new instances.

In this paper, in the wake of the most recent literature which tries to define in a better way the locality of the explanation [5–7], we present a novel *local* and *model-agnostic* XAI methodology, named *Pivot-Aided Space Transformation for Local Explanations* (PASTLE)¹, which aims to enrich the above-mentioned feature-based explanations with information about how the prediction would change if feature changes were applied. For instance, an individual whose loan request has been rejected from a bank could not only wish to understand what the main attributes which caused the rejection are, but also the things to be done in order to successfully complete the request process (e.g. applying for a loan with a lower interest rate).

Furthermore, through the computation of the relevance values of user-selected points, PASTLE can also help “expert” analysts to understand model's weaknesses and biases (e.g. when the prediction is supported by instances belonging to different classes).

We experimented our approach on a variety of classification models and diverse data sets in terms of size, dimensionality and number of classes. With the purpose of evaluating the quality of our explanation process and the enhancement of explainability with respect to standard feature-importance approaches, we compared our approach to LIME, obtaining promising results.

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¹ code is available at <https://github.com/marcopost-it/PASTLE>

example of proximity function is:

$$\rho(x_1, x_2) = \frac{1}{1 + \delta(x_1, x_2)} \quad (2)$$

whose values are in the range $[0,1]$. $\rho(x_1, x_2) = 1$ when $\delta(x_1, x_2) = 0$ (i.e. the two objects are identical) and tends to 0 when $\delta(x_1, x_2)$ approaches infinity. The effect of proximity and distance functions will be evaluated in the experimental section.

3.2. Transparent model fitting

Once neighbors are generated and labeled by the black-box predictor, two transparent classification models g', g'' (e.g. linear model, decision tree) are fitted on original and transformed data, respectively. In both cases, the aim is to approximate the decision function learned by the black-box model in the closest-range of the target instance, but their output is totally different. In particular, given the target instance t , the output of the first model is:

$$g'(t) = \sum_{i=1}^d t_i \cdot w_i, \quad (3)$$

d being the number of features, t_i being the i th component of t and w_i being the weight representing the importance of the value of the i th feature.

On the other hand, g'' learns a weight for each pivot:

$$g''(\hat{t}) = \sum_{p \in \mathbb{P}} \rho(t, p) \cdot w_p, \quad (4)$$

$\mathbb{P}$ being the set of pivots, w_p being the weight of pivot p , which represents to which extent the proximity of t to p influences the prediction.

It is worth to note that g' and g'' fitting procedures, as well as the space transformation, do not depend on the dataset size but only on the number of pivots and neighbors generated in the locality of the target instance, thus making PASTLE scalable to big data applications.

4. Explanation as a sum of forces

In a few words, the explanation flow outlined in Section 3 allows us to retrieve *feature-importance* and *pivot-importance* values. The contribution of pivots on predictions can be exploited to provide *instance-based* explanations [17,18], which are typically useful when dealing with multimedia data, such as images. For example, let's consider a classifier capable of diagnosing infection diseases based on radiography results: pivots can represent “baseline” patients with well-known symptoms, while the influence values of their proximity to an unknown condition, which the black-box model has diagnosed as “infected”, provide useful insights about the extent to which the acquired knowledge applies to the new case.

Aiming to enhance the explainability of its results even in the case of other data types (e.g. tabular, text data), PASTLE exploits pivots' influence values to understand how the black-box prediction would change if feature changes were applied to the target instance.

In this view, the output of the transparent model g'' fitted on transformed data can be seen as a sum of attractive and repulsive forces which push the target instance towards its prediction. Each pivot p exerts an attractive force if its weight w_p is positive (i.e. it positively contributes to the prediction), and vice versa. The contribute of each pivot is a function of its weight and its proximity to the target instance: decreasing the distance between the target instance and a positive-weight pivot, the likelihood of the predicted

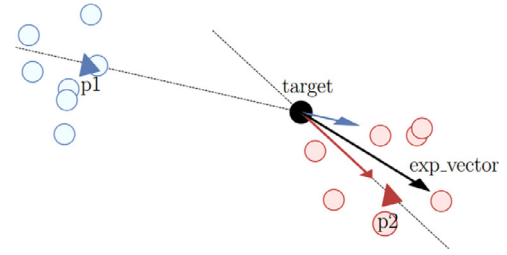


Fig. 2. Explanation vector of a target instance to which the model has assigned the red label. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

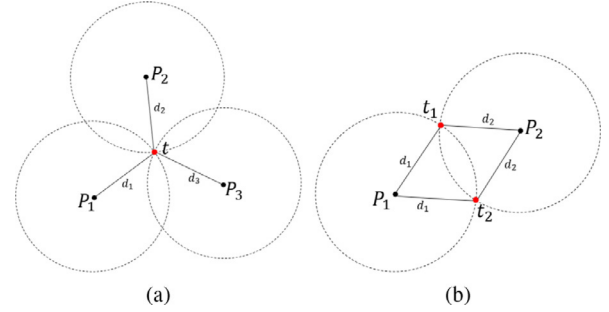


Fig. 3. (a) Bi-dimensional space with 3 pivots. If Euclidean distance were chosen as the proximity measure, the instance t would be transformed in $\hat{t} = (d_1, d_2, d_3)$. Three not-aligned pivots are enough to guarantee that $\hat{t}$ matches t and only t since it is the only intersection between the three circumferences with pivots as centers and distances as radiuses. (b) Choosing only 2 pivots, the transformed instance $\hat{t} = (d_1, d_2)$ matches two points t_1 and t_2 (the intersections of the two circumferences), leading to a possible loss of information.

class increases, and how much more the weight is big. Considering these forces as vectors and computing their sum, the resulting *explanation vector* is directed to the most supporting area for the prediction and its magnitude is the prediction $g''(\hat{t})$ itself. In addition, the opposite direction of the explanation vector provides the area along which the likelihood of the target class decreases, thus representing the opposing direction to the prediction. A simple bi-dimensional example is shown in Fig. 2: the pivot $p2$, which represents the red class, attracts the red target instance to itself – i.e., it positively contributes to the prediction –, while the pivot $p1$, representative of the blue class, exerts a repulsive force. Moving along the direction outlined by the explanation vector, points are generated in such a way that they get closer to the positively-contributing pivots, while moving away from negative ones, hence ensuring an increase of the probability of the predicted class.

5. How to choose pivots

When transforming the instance space, it is essential to ensure that information represented in the original space is preserved in the new one. From a mathematical perspective, it means we should ensure the injective property of the space transformation: every element in the transformed space is the image of at most one element of the original space. This can be accomplished if an adequate number of pivots is involved in the transformation. More specifically, it is necessary to choose at least $d + 1$ pivots (where d is the dimensionality), as shown in Fig. 3 with a simple bi-dimensional example.

The pivots' selection strategy may depend on the user who interacts with the system. On the one hand, levels of understandability required by the audience differ whether they are end-users or “expert” analysts [19]: while the former could be interested in merely understanding why the model has given its prediction (e.g. why their loan request have not been accepted), the latter could want to investigate model weaknesses to improve it (e.g. under-

Table 1
Datasets used in experimental evaluations.

Dataset	Size	# features	# classes
Bank [2]	1.371	4	2
Titanic [3]	2.200	7	2
Diabetes [2]	768	8	2
Magic [1]	19.020	10	2
Spambase [1]	4.601	57	2
Digits [1]	10.992	16	10
Breast Cancer [1]	539	30	2
Wholesale [1]	440	6	2

standing if there is any bias due to the training data); on the other hand, levels of model knowledge differ among the audiences: developers have information about the training process and can access model data, end-users have the possibility only to access to the model interface.

PASTLE allows users with training data access and information about the training process (e.g. developers, expert analysts) to deeply investigate model dynamics by analyzing both the supporting directions and the pivots' relevance values. Expert analysts with data access can indeed extract pivots in a controlled manner, exploiting clustering techniques and their domain knowledge in order to choose pivots which guarantee a good separation between classes within the transformed space. For example, several pivot-based metric indexing strategies can be properly exploited to make the pivot selection the and nearest neighbors or range query search more efficient [20].

When the end-user has access to the data but does not have the knowledge or time to manually select pivots, a random selection strategy enables an automated explanation procedure which empirically revealed to not degrade performance, as will be shown in Section 6.

In the most general case concerning end-users with no data access and no training information, we can adopt a fully automated procedure where the pivots are selected by random perturbation of the target instance. On the other hand, *expert analysts*.

6. Experiments

Aiming to evaluate the best design choices and the advantages and limitations of our algorithm, we test it on various AI models trained on real-world machine learning data sets with different size, dimensionality and number of classes, as summarized in Table 1. Specifically, we considered four classifiers: random forest (RF), support vector machine (SVM), multilayer perceptron (MLP) and XGBoost. Since the aim of this paper is not focused on building accurate classifiers, we used default configurations of the above-mentioned models, which always guarantee high accuracy values (always greater than 75%).

We assess the quality of the proposed space transformation, which turned out to guarantee higher performance (in terms of adjusted-R²) of the linear model returned as explanation has with respect to the one fitted in the original space. We evaluate the variation of the quality of the transformation to changes in the number of pivots involved and the proximity and distance functions, obtaining insights which can guide users to properly use our framework for their purpose.

Finally, we run a user study to determine the enhancements in explainability as compared with LIME (which returns the linear model fitted in the original space as explanation), achieving promising results.^{2 3}

² <https://archive.ics.uci.edu/ml/datasets/>

³ <https://www.kaggle.com/>

Table 2
Effects of the space transformation on linear model performance.

Dataset	Model	Original Space	Transformed Space
Bank	RF	0.567 ± 0.044	0.629 ± 0.025
	SVM	0.506 ± 0.033	0.571 ± 0.029
	MLP	0.550 ± 0.030	0.597 ± 0.024
	XGBoost	0.515 ± 0.028	0.561 ± 0.022
Titanic	RF	0.393 ± 0.003	0.445 ± 0.008
	SVM	0.507 ± 0.033	0.573 ± 0.029
	MLP	0.442 ± 0.020	0.493 ± 0.015
	XGBoost	0.384 ± 0.021	0.424 ± 0.018
Diabetes	RF	0.729 ± 0.033	0.765 ± 0.021
	SVM	0.753 ± 0.048	0.786 ± 0.041
	MLP	0.499 ± 0.047	0.508 ± 0.047
	XGBoost	0.592 ± 0.031	0.615 ± 0.027
Magic	RF	0.551 ± 0.028	0.614 ± 0.030
	SVM	0.667 ± 0.056	0.691 ± 0.053
	MLP	0.291 ± 0.051	0.316 ± 0.053
	XGBoost	0.362 ± 0.032	0.393 ± 0.035
Spambase	RF	0.501 ± 0.016	0.619 ± 0.004
	SVM	0.718 ± 0.041	0.688 ± 0.040
	MLP	0.391 ± 0.023	0.418 ± 0.023
	XGBoost	0.336 ± 0.026	0.435 ± 0.020
Digits	RF	0.507 ± 0.029	0.601 ± 0.026
	SVM	0.371 ± 0.032	0.628 ± 0.031
	MLP	0.378 ± 0.032	0.411 ± 0.026
	XGBoost	0.386 ± 0.024	0.460 ± 0.021

6.1. Quality of the pivot-aided transformation

Besides ensuring the injective property, we evaluated experimentally the extent to which the space-transformation affects the explanation process. Specifically, we have collected the adjusted coefficients of determination (*adjusted R²s*) of the linear model fitted on the transformed neighborhood and we compared them with the ones computed on the linear model in the original space. The *adjusted-R²* is an indicative measure of the level of explained variability in the data set and, differently than *R²*, it is independent by the number of variables which have been used to fit the linear model, allowing us to compare results in the two spaces without any bias. Note that for this evaluation we explained instances from the testing set which neither the model nor the explainer have never seen.

Considering samples of 50 measures, the average *adjusted-R²s* and their 95% confidence intervals have been collected for each data set. Results shown in Table 2 bring out that not only does the space transformation not degrade the explanation process, but it also allows to improve the goodness of the (local) linear model (the corresponding cases are outlined in bold in the table). Moreover, these results apply to all classification models we considered and do not depend by the number of features or classes the problem has.

6.2. How to choose distance and proximity functions?

The instance-space transformation which is at the core of this paper requires a *proximity function* to be computed, i.e. a function which outputs a number representing the proximity/similarity between two data points, which, in turn, is based on the computation of the *distance* between the two involved instances.

We evaluated our approach to changes in distance and proximity functions. In particular, we tested the *Euclidean*, *Minkowski*, *Chebyshev* and *cosine* distance functions and the following proximity functions:

$$P_1(x_1, x_2) = \frac{1}{1 + \delta(x_1, x_2)} \quad (5)$$

$$P_2(x_1, x_2) = e^{-\delta(x_1, x_2)} \quad (6)$$

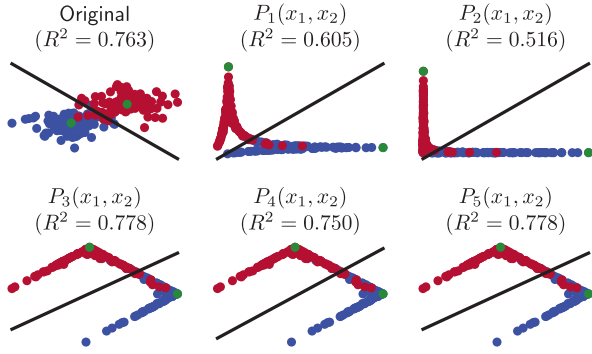


Fig. 4. Transformed spaces resulting from the choice of different proximity functions. Data from two classes (red and blue points) are reported, together with pivots (green points) and the decision boundaries of linear models, which R^2 s are reported in titles. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$$P_3(x_1, x_2) = -\delta(x_1, x_2) \quad (7)$$

$$P_4(x_1, x_2) = 1 - \frac{\delta(x_1, x_2) - \min(\delta(x_1, x_2))}{\max(\delta(x_1, x_2)) - \min(\delta(x_1, x_2))} \quad (8)$$

$$P_5(x_1, x_2) = \max(\delta(x_1, x_2)) - \delta(x_1, x_2), \quad (9)$$

x_1 and x_2 being the two data points, $\delta(x_1, x_2)$ representing a generic distance function and $\min(\delta(x_1, x_2))$ and $\max(\delta(x_1, x_2))$ representing the minimum and maximum distance observed in the whole dataset, respectively.

Figure 4 shows the different transformed spaces generated by choosing two pivots and the above-defined proximity functions. What stands out in the figure is that distortions caused by non-linear proximity functions result in a worse linear model fitting, while the others can even improve it.

Consistently with the insights described above, results in Table 3 make clear that the best proximity functions are the ones shown in Eqs. (7), (9), where the proximity has a linear dependence with the distance. By making the distance function $\delta(x_1, x_2)$ vary, we can conclude that cosine distance achieves the best performance on average, but the best configurations usually involve Euclidean or Minkowski distance functions, which guarantee better results when choosing proximity functions which are linearly dependent with the distance.

6.3. How does the quality vary by changing the number of pivots?

Figure 5 shows the adjusted R^2 s varying the number of pivots involved in the transformation. The empirical results confirm our theoretical analysis according to which it is necessary to choose a number of pivots greater than the number of attributes of the instances to explain. However, in certain cases, the performance of the original space is surpassed even with a smaller number of pivots.

6.4. Does the pivot extraction process affect quality?

As detailed in the previous section, the pivots' selection strategy depends on the user who is interacting with the system and by the data access privileges. We experimented the proposed strategies by considering samples of 50 instances and measuring the adjusted- R^2 s and their 95% confidence intervals in the transformed space. Formally, we conducted a paired t-test to assess whether the space transformation improves performance. The *perturbation-based* strategies have been tested considering for each feature

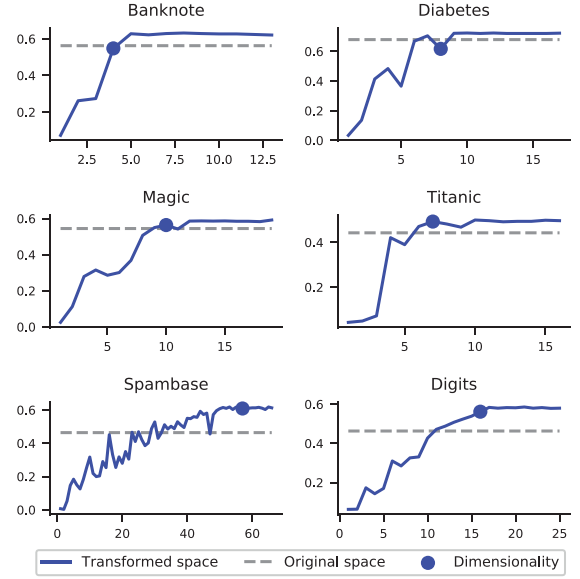


Fig. 5. Effects of the number of pivots.

the ranges $[v - p \cdot v, v + p \cdot v]$, v being the target instance feature's value and $p \in \{0.5, 1, 2\}$; the *Random* and *Clustering* strategies have been implemented by sampling the training set instances with an uniform distribution and performing the KMeans algorithm over the training set, respectively.

Results of our evaluations are shown in Table 4. Bold values indicate that the result is statistically equivalent or better than the result obtained in the original space, while underlined values represent the best solution. *Clustering* and *Random* selection strategies reveal them to be very effective, improving the original-space results for each dataset. The *Perturbation* approach, despite being sufficient in some cases, reveals its weaknesses: for example, perturbations on the Spambase dataset – which features represent the frequency of words in emails – 8 do not perturb zeros, thus not inserting new words but only changing the frequency of those already existing, resulting in a poor space transformation.

6.5. Does the end-user benefit from PASTLE explanation?

We base our evaluation of explainability on the concept proposed by Barredo Arrieta et al. [19], which defines it as “the ability of a model to make its functioning clearer to an audience”. The audience considered in our user study is composed of 36 users with basic knowledge about Machine Learning. The aim of our study is to quantify the extent to which PASTLE explanations for a few instances help users understand how the decision model works in unknown situations.

Since our aim was to evaluate the improvement implied by the directions provided by PASTLE over the attribute-relevance values provided by feature importance approaches, we based our study on two specular surveys intended for two separate and independent halves of the audience: on the basis of two classification models on the *Breast Cancer Wisconsin* [1] data set (M_1) and the *Wholesale Customer* [1] dataset (M_2), the first survey shows PASTLE explanations for M_1 and LIME explanations for M_2 , while the second one shows LIME explanations for M_1 and PASTLE explanations for M_2 .

The audience, after having seen explanations for 5 instances, had to sort them in ascending order according to the probability of a predetermined class (“Horeca” for Wholesale Customer dataset, “Benign” for Breast Cancer Wisconsin dataset) assigned by the related classification model.

Table 3
Effects of distance and proximity functions.

	Proximity	Distance				mean
		Euclidean	Minkowski	Chebyshev	Cosine	
Bank	P_1	0.227	0.279	0.25	0.382	0.284
	P_2	0.216	0.215	0.158	0.571	0.29
	P_3	0.689	0.683	0.412	0.684	0.617
	P_4	0.33	0.333	0.26	0.648	0.393
	P_5	0.686	0.686	0.392	0.685	0.612
	mean	0.43	0.439	0.294	0.594	
	P_1	0.278	0.278	0.162	0.452	0.292
	P_2	0.152	0.157	0.141	0.428	0.219
	P_3	0.512	0.517	0.176	0.511	0.429
	P_4	0.395	0.395	0.139	0.436	0.341
Titanic	P_5	0.5	0.5	0.18	0.501	0.42
	mean	0.367	0.369	0.161	0.466	
	P_1	0.312	0.315	0.209	0.637	0.368
	P_2	0.146	0.148	0.199	0.639	0.283
	P_3	0.701	0.702	0.216	0.718	0.584
	P_4	0.584	0.584	0.268	0.707	0.536
	P_5	0.692	0.694	0.333	0.738	0.614
	mean	0.487	0.489	0.245	0.634	
	P_1	0.136	0.138	0.225	0.436	0.234
	P_2	0.084	0.087	0.155	0.444	0.192
Diabetes	P_3	0.581	0.586	0.15	0.563	0.47
	P_4	0.469	0.469	0.31	0.555	0.451
	P_5	0.576	0.574	0.147	0.554	0.463
	mean	0.369	0.371	0.197	0.51	
	P_1	−0.045	−0.045	0.007	0.254	0.043
	P_2	−0.061	−0.061	−0.042	0.368	0.051
	P_3	0.582	0.581	0.197	0.507	0.467
	P_4	0.44	0.437	0.123	0.534	0.383
	P_5	0.576	0.58	0.221	0.511	0.472
	mean	0.298	0.298	0.1	0.435	
Magic	P_1	0.078	0.078	0.027	0.278	0.115
	P_2	−0.001	−0.002	0.039	0.411	0.112
	P_3	0.563	0.56	0.097	0.509	0.432
	P_4	0.451	0.453	0.07	0.497	0.368
	P_5	0.559	0.565	0.086	0.513	0.431
	mean	0.33	0.331	0.064	0.442	
	P_1	0.078	0.078	0.027	0.278	0.115
	P_2	−0.001	−0.002	0.039	0.411	0.112
	P_3	0.563	0.56	0.097	0.509	0.432
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	P_2	−0.001	−0.002	0.039	0.411	0.112
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	P_4	0.451	0.453	0.07	0.497	0.368
	P_5	0.559	0.565	0.086	0.513	0.431
	mean	0.33	0.331	0.064	0.442	
	P_1	0.078	0.078	0.027	0.278	0.115
	P_2	−0.001	−0.002	0.039	0.411	0.112
Digits	P_3	0.563	0.56	0.097	0.509	0.432
	P_4	0.451	0.453	0.07	0.497	0.368
	P_5	0.559	0.565	0.086	0.513	0.431
	mean	0.33	0.331	0.064	0.442	
	P_1	0.078	0.078	0.027	0.278	0.115
	P_2	−0.001	−0.002	0.039	0.411	0.112
	P_3	0.563	0.56	0.097	0.509	0.432
	P_4	0.451	0.453	0.07	0.497	0.368
	P_5	0.559	0.565	0.086	0.513	0.431
	mean	0.33	0.331	0.064	0.442	

Table 4
Effect of the pivots' generation strategy.

Dataset	Perturb. (0.5)	Perturb. (1)	Perturb. (2)	Random	Clustering
Bank	0.5 ± 0.034	0.555 ± 0.034	0.571 ± 0.032	0.592 ± 0.034	0.65 ± 0.017
Titanic	0.372 ± 0.02	0.385 ± 0.023	0.394 ± 0.022	0.484 ± 0.025	0.495 ± 0.014
Diabetes	0.704 ± 0.025	0.706 ± 0.028	0.702 ± 0.023	0.709 ± 0.022	0.728 ± 0.019
Magic	0.381 ± 0.038	0.396 ± 0.035	0.421 ± 0.041	0.545 ± 0.027	0.545 ± 0.035
Spambase	0.068 ± 0.024	0.086 ± 0.026	0.092 ± 0.026	0.586 ± 0.008	0.568 ± 0.008
Digits	0.449 ± 0.03	0.448 ± 0.028	0.448 ± 0.029	0.547 ± 0.026	0.559 ± 0.026

Table 5
Survey results: P@k and mAP.

Metric	Wholesale		Breast Cancer	
	LIME	PASTLE	LIME	PASTLE
mAP	0.2014	0.3013	0.1127	0.3569
P@3	0.2269	0.2970	0.1155	0.3919
P@5	0.250	0.250	0.170	0.325

6.5.1. Mean precision and mean average precision

The collected results have been used to compute *Mean Precision* ($P@k$) and *Mean Average Precision* (*mAP*) metrics. $P@k$ is the fraction of correctly sorted instances among the first k of the ranking: we measured it considering just the first three instances ($P@3$) and all instances ($P@5$). *Mean Average Precision* penalizes errors in the first positions of the ranking averaging the overall *Average Precision* (AP) value obtained for each answer.

The results shown in Table 5 prove that PASTLE always achieves better performance than LIME, especially for the Breast Cancer

problem where features are far less interpretable, thus confirming our hypothesis that the provided supporting and opposing directions help end-users to understand the model's decisions.

6.5.2. Weighted displacements

For this experiment, we have slightly modified the test proposed by Santini [21] to quantify the difference between two performed rankings (a.k.a. treatments) in terms of hypothesis verification. Let I be the set of n instances shown to the user and c the reference class for the experiment. There is a ground truth ordering based on the probability of the instances to be assigned to class c by the black-box model, represented as $L_t = \{I_1, \dots, I_n\}$. Every instance i in the ordering is associated to a measure of relevance $0 \leq S(I_i) \leq 1$ such that $S(I_i) \geq S(I_{i+1})$ for the ground truth. In our case, the measure of relevance is defined as the probability of class c assigned by the black box model M , i.e. $S(I_i) = P_M(y = c | I_i)$. This is compared with an experimental ordering guessed by users after having seen explanations $L_d = \{I_{\pi_1}, \dots, I_{\pi_n}\}$ where $\{\pi_1, \dots, \pi_n\}$ is a permutation of $1, \dots, n$. The displacement of I_i is defined as $d(I_i) = |i - \pi_i|$, while the relative displacement of L_d is computed

Table 6
Survey results: weighted displacements.

	Wholesale		Breast Cancer	
	LIME	PASTLE	LIME	PASTLE
μ	1.112	1.029	1.247	1.034
σ^2	0.067	0.193	0.056	0.161
F		0.410		3.686

as follows:

$$W = \frac{\sum_i S(l_i) d(l_i)}{\Omega}, \quad (10)$$

$\Omega = \lfloor \frac{n^2}{2} \rfloor$ being a normalization factor.

In our experiments, rankings provided by users after having seen explanations provided by LIME and PASTLE have been compared with the ground truth with the aim to determine whether the observed differences can be ascribed to the different explanation technique or to random variation. Hence, the null hypothesis H_0 is that all the average scores obtained with all the treatments are the same, and we test its acceptance with the F ratio. In our experiments, we have $m=2$ treatments (PASTLE and LIME rankings) and $n=18$ measurements (user answers). Let w_{ij} be the result of the j th experiment on the i th treatment. Define $\mu_i = \frac{1}{n} \sum_{j=1}^n w_{ij}$ as the average result for the i th treatment, $\mu = \frac{1}{m} \sum_{i=1}^m \mu_i$ as the total average, $\sigma^2 = \frac{n}{m-1} \sum_{i=1}^m (\mu_i - \mu)^2$ as the variance *between* treatments and $\sigma_W^2 = \frac{1}{m(n-1)} \sum_{i=1}^m \sum_{j=1}^n (w_{ij} - \mu_i)^2$ as the variance *within* treatments. Then, the F ratio is computed as $F = \frac{\sigma^2}{\sigma_W^2}$: a high value indicates that the differences in the averages are likely to be due to treatments. As can be noted from results reported in Table 6, the null hypothesis H_0 is rejected (in favour of PASTLE) in the case of the *Breast Cancer* dataset. On the other hand, no significant differences have been found between mean scores of PASTLE and LIME in the case of the *Wholesale* dataset. This is probably due to a greater simplicity of the task resulting from a higher interpretability of dataset features (e.g. annual spending on fresh, milk and grocery products to classify the kind of wholesaler client, i.e. retailer or horeca).

6.6. What are the practical uses of PASTLE?

After having proved the effectiveness of PASTLE, we point out its applicability to various machine learning's problems. Specifically, our framework, besides being model-agnostic, can be plugged in any classification and regression pipelines which require an explanation in the form of features' importance.

6.6.1. Classification

Figure 6 (a) depicts an example of explanation for a diabetic patient in the *Pima Indians dataset* [2]: the green (red) region contains features which positively (negatively) support the predicted class; specifically, the position on the horizontal axis is determined according to the features importance learnt by g' : for instance, *Glucose* is the feature which most-positively contributes to the prediction because of the highest weight that g' associated to it. In addition, the arrows indicate the direction in which the corresponding feature should be changed in order to have the same prediction with higher probability; note that the changing direction is the explanation vector's relative component computed considering the target instance as the origin. In this case, for instance, *Glucose* is the most important feature supporting the prediction and, together with *Insulin* and *BloodPressure*, should be increased in order to support the "Diabetes" prediction.

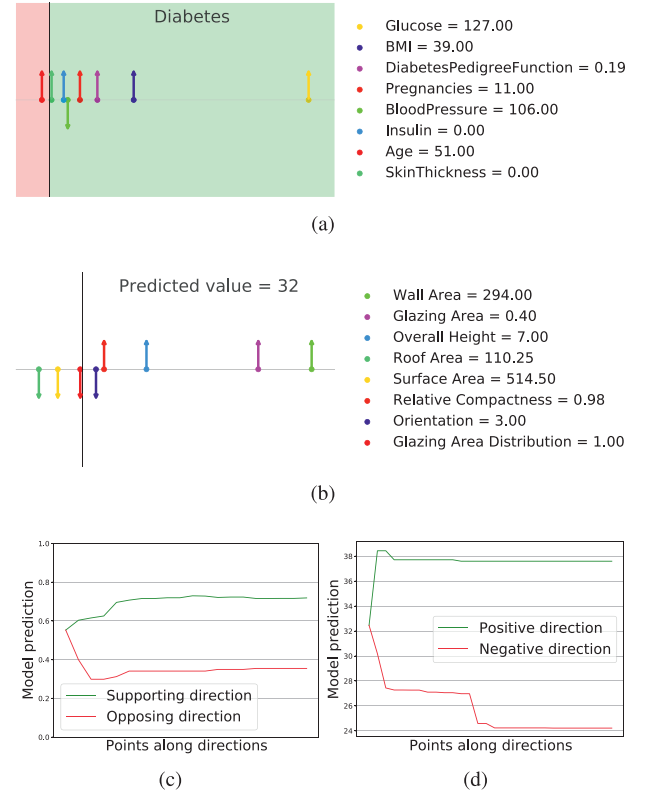


Fig. 6. Example of a diabetic patient: (a) explanation and (b) how the probability of the disease changes along supporting and opposing directions.

The *explanation vector* allows to understand how the prediction changes when changing feature values: moving towards the supporting direction, the likelihood of the predicted class increases, and vice versa. Figure 6(c) shows an example considering artificial points generated along those directions.

6.6.2. Regression

When considering regression problems, the numeric output is not discrete and the explanation should only suggest the direction in which each feature should be modified in order to increase/decrease the predicted value. Figure 6(b) shows an example for the Energy Efficiency Dataset [1] whose aim is to predict the heating load of (simulated) buildings: while the top right position of the features *wall area*, *glazing area* and *overall height* reveals they positively contributed to the prediction, their corresponding upper oriented arrows suggests to increase their values to amplify the heating load of the building. In addition, as for the classification problem, Fig. 6(d) shows how the prediction would change following the direction provided by the *explanation vector*, confirming that PASTLE explanation really describes what the model has learnt in the locality of the target instance.

7. Conclusion

In this paper, we propose a novel local model-agnostic XAI approach, named PASTLE, which enriches standard "feature importance" explanations with information about the changes which have to be applied to the target instance in order to increase (or decrease) the probability of the label assigned by the model. The experiments on various data sets and the user study verify the usefulness of PASTLE.

In future works, we would apply our framework to different problems, with the aim to evaluate its practical usefulness, especially in Big Data applications. Furthermore, we would extend it

to receive different kinds of data (e.g. images, graphs) and models (e.g. regression, recurrent models) as inputs.

Declaration of Competing Interest

None.

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